SAKARYA ÜNİVERSİTESİ

# Inferential Statistical Analysis of Hospital Length of Stay: Extending the Analytical Perspective of Inpatient Fall Prediction Research

Y255052090 — Hanifullah Stanikzai

Supervisor: Prof. Dr. İSMAİL HAKKI CEDİMOĞLU

Course Name: ÇIKARIMSAL İSTATİSTİK

Bilişim Sistemleri Mühendisliği

# The Problem: Inpatient Falls

### Patient Safety Crisis

Falls cause injuries, extended hospitalization, and rising costs

### ML-Dominated Research

Most studies optimize AUC and F1-Score over statistical interpretation

### The Gap

Prediction accuracy alone cannot quantify uncertainty or variable relationships

# Foundational Study

SELECTED PAPER

## "Machine Learning Models Predicting Inpatient Falls"

Develops and evaluates ML models for inpatient fall prediction

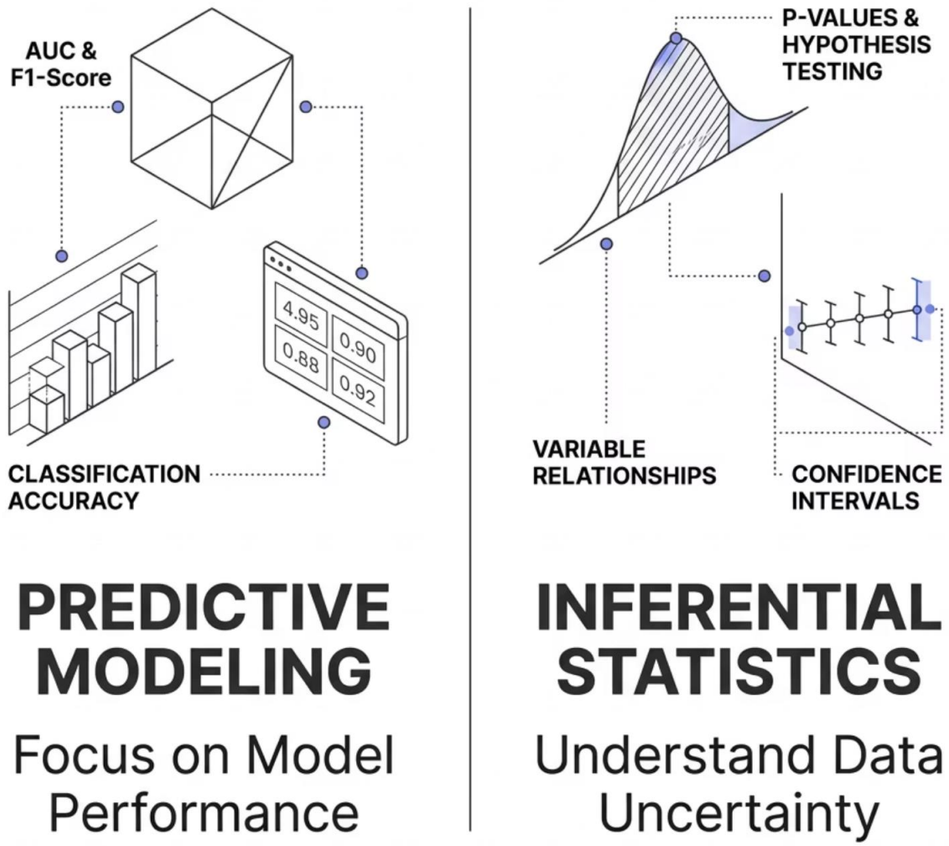
**Limitation:** Emphasis on predictive performance; minimal inferential statistical analysis

AUC · F1-Score · Sensitivity · Specificity

# Identifying the Research Gap

Healthcare prediction literature prioritizes **model performance metrics** over statistical inference — leaving uncertainty and variable relationships underexplored.

**Core Question:** How can inferential statistical techniques complement predictive models in healthcare data analysis?



# Our Analytical Contribution

Applying four inferential methods to complement ML-based fall prediction research

### Bootstrap Sampling

10,000 resamples to estimate sampling distributions

### Confidence Intervals

95% CI for mean length of stay

### Hypothesis Testing

Two-sample t-test comparing LOS across admission types

### Linear Regression

Modeling admission type–LOS relationships

# Dataset: MIMIC Clinical Database

Original study data was unavailable. Analysis used a **hospital admission dataset derived from MIMIC.**

**Key variables:** Admission Type · Insurance · Ethnicity · Length of Stay (LOS)

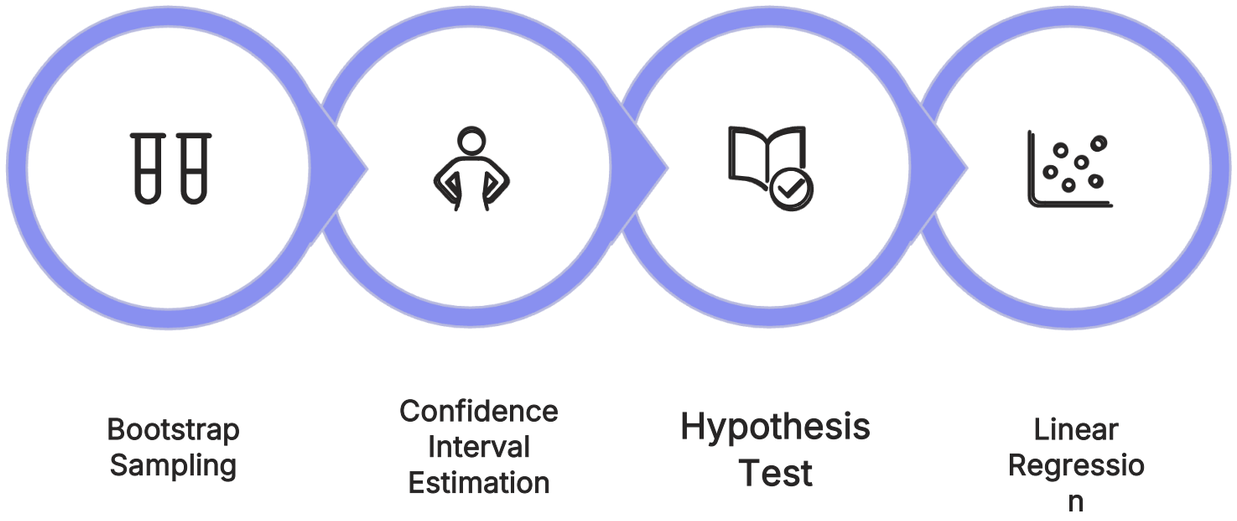
**129**

Hospital admissions

**20**

Variables

# Statistical Methods Implemented



# Key Results

**9.34**

**Mean LOS (days)**

Bootstrap-estimated average

**0.588**

**p-value (t-test)**

No significant LOS difference by admission type

**0.604**

**p-value (regression)**

Admission type explains minimal LOS variability

95% CI for mean LOS: **7.46 – 11.82 days** · t-statistic = **−0.56**

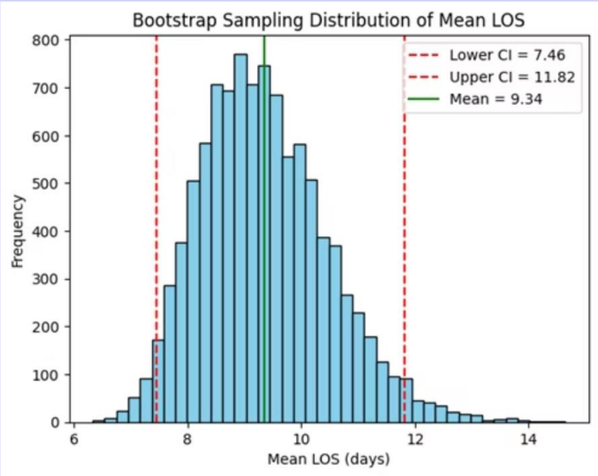



**Fig. 1.** Bootstrap distribution of hospital length of stay using 10,000 resamples

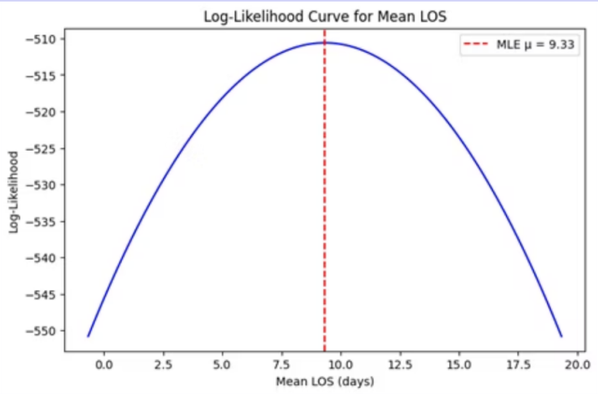



**Fig. 2.** Log-likelihood curve for the normal distribution parameters of hospital length of stay.

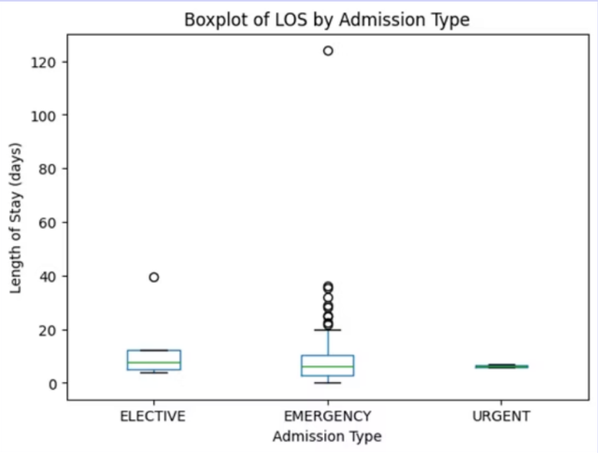



**Fig. 4.** Boxplot comparison of hospital length of stay between emergency and elective admissions.

# Discussion

## Bootstrap Validated

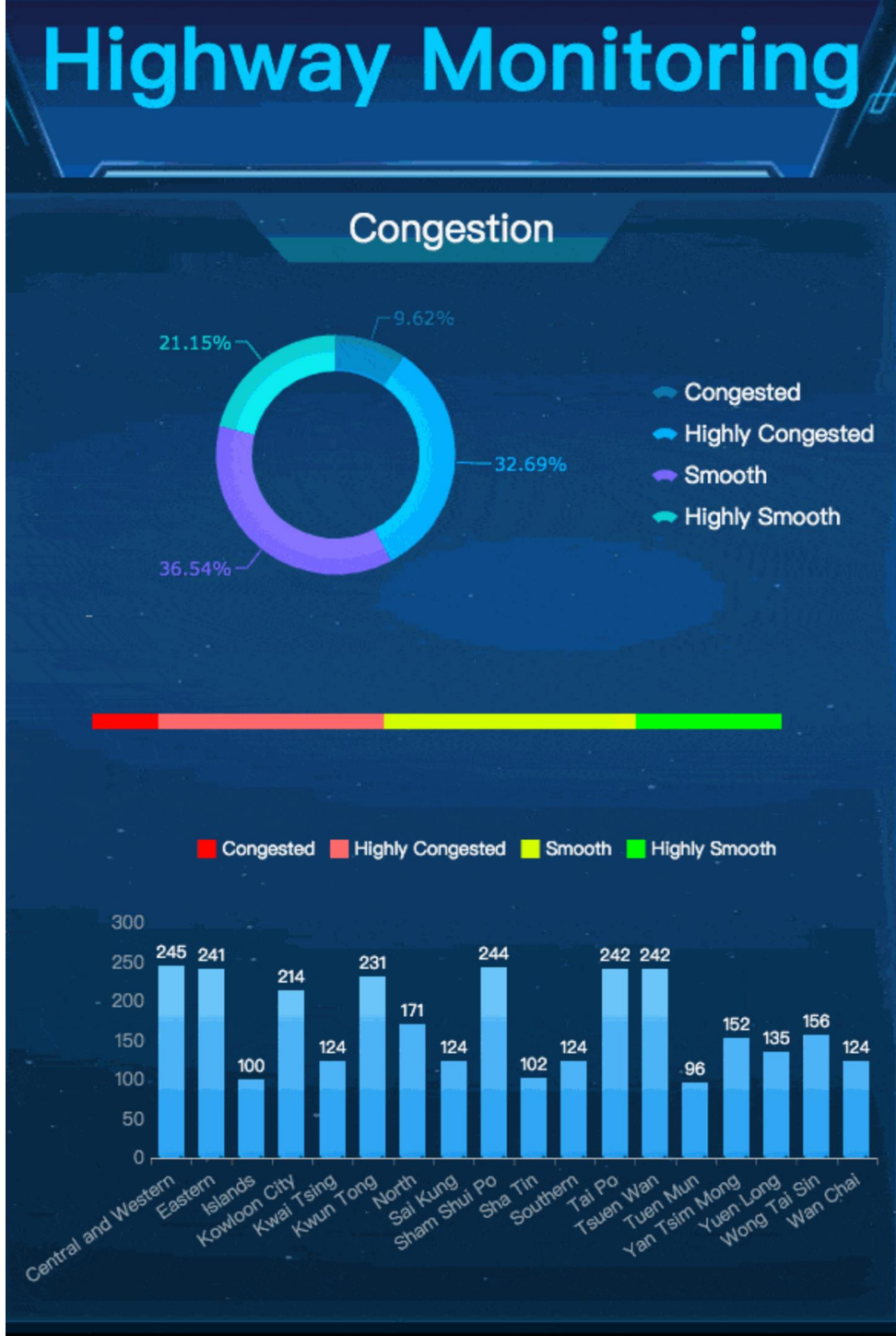
Sampling distribution and 95% CI successfully constructed for LOS

## No Significant Association

Admission type did not significantly predict LOS (p = 0.588)

## Broader Implication

LOS likely driven by clinical or demographic variables beyond admission type



# Conclusion

- Four inferential methods applied to MIMIC-derived hospital data
- No significant admission type–LOS relationship detected
- Results constrained by sample size (n=129) and data availability
- Demonstrates value of statistical inference in healthcare analytics

**Future direction:** Expand sample size; incorporate clinical and demographic predictors